AI-Based Urban Climate Resilience Score Using Multimodal Environmental Intelligence

Problem Statement

Climate change is increasing the frequency of heatwaves, urban flooding, poor air quality, and loss of green spaces. While cities collect data from weather stations, air quality sensors, satellite images, and IoT devices, these datasets are rarely integrated into a single intelligent system that can evaluate a city's resilience to climate change.

Current systems monitor individual parameters such as temperature or pollution but do not provide a comprehensive assessment of how prepared different city regions are to withstand climate-related events.

The proposed research aims to develop an **AI-Based Urban Climate Resilience Score** that integrates environmental, climatic, and urban infrastructure data to evaluate the resilience of different city zones. The system will identify climate-vulnerable areas, predict future climate risks, and recommend suitable adaptation strategies such as increasing green cover, improving drainage systems, and reducing urban heat islands.

Objectives

1. Develop an AI model to compute an Urban Climate Resilience Score (0–100).
2. Integrate satellite imagery, IoT sensors, weather data, and air quality measurements.
3. Predict heatwave, flood, and pollution vulnerability.
4. Generate explainable recommendations for climate adaptation.
5. Visualize resilience maps for city planners.
6. Evaluate the effectiveness of proposed interventions.

Public Datasets

Dataset	Purpose	Link
Copernicus Sentinel-2	Land cover, vegetation, urban expansion	https://dataspace.copernicus.eu/
NASA POWER	Weather, solar radiation, temperature	https://power.larc.nasa.gov/

OpenAQ	Air quality (PM2.5, PM10, NO_2, O_3)	https://openaq.org/
WorldClim	Climate variables	https://www.worldclim.org/
Global Surface Water Explorer	Water bodies and flood analysis	https://global-surface-water.appspot.com/
OpenStreetMap	Roads, buildings, green spaces	https://www.openstreetmap.org/
ERA5 Climate Reanalysis	Historical climate data	https://cds.climate.copernicus.eu/

Expected Outcomes

- Urban Climate Resilience Score for each city zone.
- Climate vulnerability maps.
- AI-based adaptation recommendations.
- Decision-support dashboard for urban planners.
- Prediction of future climate risks.